

Math 1232 Summer 2026

Practice Final

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, **two-sided** cheat sheet you have made for yourself ahead of time.
- You may use a four-function calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- Each problem is worth ten points. The whole test is scored out of 100 points.
- The exam has 14 problems, the first 8 of which you must do.
- You must solve at least 2 of the last 6 problems. If you solve more than 2, your best 2 scores will be recorded for points as problems 9 and 10, and any additional (successful) solutions will count for a bonus point or two.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

1	2	3	4	5
6	7	8	9	10

Total /100

Name: **Solutions**

Problem 1. Find the general solution to the equation $y' = e^{x+y}$.

Solution.

$$\begin{aligned}\frac{dy}{dx} &= e^x e^y \\ e^{-y} dy &= e^x dx \\ -e^{-y} &= e^x + C \\ -y &= \ln(-e^x - C) \\ y &= -\ln(-e^x - C).\end{aligned}$$

Problem 2. Let $b_n = \frac{n! - 2}{(n+2)!}$. Compute $\lim_{n \rightarrow \infty} b_n$ with justification.

Solution. We could split this into two pieces and say that

$$\lim_{n \rightarrow \infty} \frac{n! - 2}{(n+2)!} = \lim_{n \rightarrow \infty} \frac{n!}{(n+2)!} - \lim_{n \rightarrow \infty} \frac{2}{(n+2)!} = \lim_{n \rightarrow \infty} \frac{1}{(n+2)(n+1)} - \lim_{n \rightarrow \infty} \frac{2}{(n+2)!} = 0$$

since both limits are zero. Alternatively, we could say that

$$0 \leq \frac{n! - 2}{(n+2)!} \leq \frac{n!}{(n+2)!} = \frac{1}{(n+2)(n+1)}$$

and since the limits of the outer sequences are zero, so is the limit of the inner sequence by the squeeze theorem.

Problem 3. Compute $\sum_{n=0}^{\infty} \ln \left(\frac{n+1}{n+3} \right)$.

Solution. We can rewrite the sequence we are summing as $\ln(n+1) - \ln(n+3)$, so the partial sums are

$$\begin{aligned}s_k &= \ln(1) - \ln(3) + \ln(2) - \ln(4) + \ln(3) - \ln(5) + \ln(4) - \ln(6) + \ln(5) - \ln(7) \\ &\quad + \cdots + \ln(k-1) - \ln(k+1) + \ln(k) - \ln(k+2) + \ln(k+1) - \ln(k+3) \\ &= \ln(1) + \ln(2) - \ln(k+2) - \ln(k+3).\end{aligned}$$

Since $\ln(x)$ grows without bound, we have that

$$\sum_{n=0}^{\infty} \ln \left(\frac{n+1}{n+3} \right) = \lim_{k \rightarrow \infty} s_k = -\infty.$$

Problem 4. Analyze the convergence of $\sum_{n=2}^{\infty} \frac{3n^3 - 2n^2 + n}{4n^6 + 4n^3 - n}$.

Solution. Here it would be hard to use the direct comparison test. Instead, we limit compare to $\frac{1}{n^3}$ since the difference in the degrees in the numerator and denominator is 3. We have

$$\lim_{n \rightarrow \infty} \frac{\frac{3n^3 - 2n^2 + n}{4n^6 + 4n^3 - n}}{\frac{1}{n^3}} = \lim_{n \rightarrow \infty} \frac{3n^6 - 2n^5 + n^4}{4n^6 + 4n^3 - n} = 3/4.$$

Since this is a finite non-zero number, the two series have the same convergence behavior. Thus, since $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges by the p -test, we know that our series also converges by the Limit Comparison Test. Since this is a series of all positive terms, the convergence is automatically absolute.

Problem 5. Analyze the convergence of $\sum_{n=2}^{\infty} (-1)^n \frac{n^3 - n}{3n^4 + n^2 + 2}$.

Solution. First we test absolute convergence. We use the limit comparison test:

$$\lim_{n \rightarrow \infty} \frac{\frac{(n^3 - n)}{(3n^4 + n^2 + 2)}}{1/n} = \lim_{n \rightarrow \infty} \frac{n^4 - n}{3n^4 + n^2 + 2} = \frac{1}{3}.$$

This is a non-zero finite limit, so the series have the same behavior. Since $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges (harmonic series or p -test with $p = 1$), we know that $\sum_{n=2}^{\infty} \frac{n^3 - n}{3n^4 + n^2 + 2}$ diverges and so $\sum_{n=2}^{\infty} (-1)^n \frac{n^3 - n}{3n^4 + n^2 + 2}$ does not converge absolutely.

But

$$\lim_{n \rightarrow \infty} \frac{n^3 - n}{3n^4 + n^2 + 2} = 0$$

and the sequence is decreasing, so by the alternating series test, our series converges conditionally.

Problem 6. Find the radius of convergence and the interval of convergence of $\sum_{n=1}^{\infty} \frac{4^n}{n^2 + n} x^n$.

Solution. We use the ratio test.

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{4^{n+1} x^{n+1}}{(n+1)^2 + n + 1}}{\frac{4^n x^n}{n^2 + n}} \right| = \lim_{n \rightarrow \infty} 4|x| \frac{n^2 + 3n + 2}{n^2 + n} = 4|x|.$$

So we need $4|x| < 1$ or $-1 < 4x < 1$, or $-1/4 < x < 1/4$. We need to have x in the interval $(-1/4, 1/4)$, so the radius is $1/4$.

To finish finding the interval we need to check the endpoints. We see

$$\sum_{n=1}^{\infty} \frac{4^n}{n^2 + n} (1/4)^n = \sum_{n=1}^{\infty} \frac{1}{n^2 + n}$$

converges by comparison to a p -series with $p = 2$

$$\sum_{n=1}^{\infty} \frac{4^n}{n^2 + n} (-1/4)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + n}$$

converges by the alternating series test or a comparison test

Thus the interval of convergence is $[-1/4, 1/4]$.

Problem 7. Write down a formula for the Taylor series of $(1 - 2x)^{-3}$, and then write down the third degree polynomial $T_3(x)$ explicitly.

Solution. We can take this from the binomial series. So we have

$$f(x) = \sum_{n=0}^{\infty} \binom{-3}{n} (-2x)^n = \sum_{n=0}^{\infty} \binom{-3}{n} (-2)^n x^n$$

$$\begin{aligned} T_3(x) &= 1 + (-3)(-2)x + \frac{(-3)(-4)}{2!}(4)x^2 + \frac{(-3)(-4)(-5)}{3!}(-8)x^3 \\ &= 1 + 6x + 24x^2 + 80x^3. \end{aligned}$$

Problem 8. Do one of the following integrals:

$$(a) \int \frac{3x^2 - 7x + 8}{x(x-2)^2} dx, \quad (b) \int x^3 \sqrt{1-x^2} dx, \quad (c) \int \arcsin(x) dx.$$

Solution. (a) We set up

$$\frac{3x^2 - 7x + 8}{x(x-2)^2} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$$3x^2 - 7x + 8 = A(x-2)^2 + B(x-2)x + Cx$$

$$x = 0 : \quad 8 = A(-2)^2 = 4A \quad \Rightarrow A = 2$$

$$x = 2 : \quad 6 = 2x \quad \Rightarrow C = 3$$

$$x = 1 : \quad 4 = 2 - B + 3 \quad \Rightarrow B = 1.$$

So we have

$$\begin{aligned}\int \frac{3x^2 - 7x + 8}{x(x-2)^2} dx &= \int \frac{2}{x} + \frac{1}{x-2} + \frac{3}{(x-2)^2} dx \\ &= 2 \ln |x| + \ln |x-2| - \frac{3}{x-2} + C.\end{aligned}$$

(b) We're going to set $x = \sin \theta$ so that $dx = \cos \theta d\theta$. Then

$$\begin{aligned}\int x^3 \sqrt{1-x^2} dx &= \int \sin^3(\theta) \sqrt{1-\sin^2(\theta)} \cos(\theta) d\theta \\ &= \int \sin^3(\theta) \cos^2(\theta) d\theta \\ &= \int \cos^2(\theta) \sin(\theta) (1 - \cos^2(\theta)) d\theta\end{aligned}$$

At this point we can take $u = \cos(\theta)$ so that $du = -\sin(\theta) d\theta$, and we get

$$\begin{aligned}\int x^3 \sqrt{1-x^2} dx &= - \int u^2 (1-u^2) du = \int -u^2 + u^4 du \\ &= \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{\cos^5(\theta)}{5} - \frac{\cos^3(\theta)}{3} + C.\end{aligned}$$

But we know that $\sin(\theta) = x$, so $\cos(\theta) = \sqrt{1-x^2}$, and we have any of the following acceptable answers:

$$\int x^3 \sqrt{1-x^2} dx = \frac{(\sqrt{1-x^2})^5}{5} - \frac{(\sqrt{1-x^2})^3}{3} + C$$

(c) We can set $u = \arcsin(x)$ and $dv = 1$ to get $du = 1/\sqrt{1-x^2}$ and $v = x$. Then

$$\begin{aligned}\int \arcsin(x) &= x \arcsin(x) - \int \frac{x}{\sqrt{1-x^2}} dx \\ &= x \arcsin(x) - \int \frac{-1}{2\sqrt{u}} du = x \arcsin(x) + \sqrt{u} + C \\ &= x \arcsin(x) + \sqrt{1-x^2} + C.\end{aligned}$$

Name: [Solutions](#)

Do at least 2 of the following 6 problems. If you successfully solve more than 2, you will receive one or two bonus points on the exam, but you should make sure to have at least something written down for each of the first 8 problems before attempting more than 2 problems in this section.

Problem 9. Is $f(x) = e^{x^3+1}$ invertible or not? Justify your answer. If it's invertible, give a formula for the inverse.

Solution. We can start by plugging in some small values, but we won't find any having the same output. Thus we should suspect it's invertible, and try to find a formula for the inverse:

$$\begin{aligned}y &= e^{x^3+1} \\ \ln(y) &= x^3 + 1 \\ \ln(y) - 1 &= x^3 \\ x &= \sqrt[3]{\ln(y) - 1}.\end{aligned}$$

So f is indeed invertible and $f^{-1}(x) = \sqrt[3]{\ln(x) - 1}$.

Problem 10. Compute $\lim_{x \rightarrow 0} \frac{7^x - 5^x}{x}$.

Solution. The numerator and denominator are both approaching 0 as x approaches 0, so we can use l'Hôpital's rule:

$$\lim_{x \rightarrow 0} \frac{7^x - 5^x}{x} = \lim_{x \rightarrow 0} \frac{\ln(7)7^x - \ln(5)5^x}{1} = \ln(7) - \ln(5).$$

Problem 11. Compute $\int_0^7 \frac{1}{\sqrt[3]{7-x}} dx$.

Solution. We know that $\frac{1}{\sqrt[3]{7-x}}$ is undefined at 7. So we have

$$\begin{aligned}\int_0^7 \frac{1}{\sqrt[3]{7-x}} dx &= \lim_{t \rightarrow 7} \int_0^t \frac{1}{\sqrt[3]{7-x}} dx \\ &= \lim_{t \rightarrow 7} -\frac{3}{2}(7-x)^{2/3} \Big|_0^t \\ &= \lim_{t \rightarrow 7} -0 + \frac{3}{2}(7)^{2/3} = \frac{3}{2}7^{2/3}.\end{aligned}$$

Problem 12. Compute the arc length of the curve $y = \frac{1}{3}(x^2 - 2)^{3/2}$ for $3 \leq x \leq 6$.

Solution. We have $y' = \frac{1}{3} \cdot \frac{3}{2}(x^2 - 2)^{1/2} \cdot 2x = x\sqrt{x^2 - 2}$. So the arc length is

$$\begin{aligned} L &= \int_3^6 \sqrt{1 + y'^2} \, dx = \int_3^6 \sqrt{1 + x^2(x^2 - 2)} \, dx \\ &= \int_3^6 \sqrt{1 - 2x^2 + x^4} \, dx = \int_3^6 \sqrt{(x^2 - 1)^2} \, dx \\ &= \int_3^6 (x^2 - 1) \, dx = \left. \frac{x^3}{3} - x \right|_3^6 \\ &= (72 - 6) - (9 - 3) = 60. \end{aligned}$$

Problem 13. Use a Taylor series to compute $\lim_{x \rightarrow 0} \frac{\cos(x^2) - 1 + x^4/2}{x^8}$.

Solution.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos(x^2) - 1 + x^4/2}{x^8} &= \lim_{x \rightarrow 0} \frac{(1 - x^4/2 + x^8/4! - x^{12}/6! + \dots) - 1 + x^4/2}{x^8} \\ &= \lim_{x \rightarrow 0} \frac{x^8/4! - x^{12}/6! + \dots}{x^8} \\ &= \lim_{x \rightarrow 0} \frac{1}{4!} - \frac{x^4}{6!} + \dots = \frac{1}{24}. \end{aligned}$$

Problem 14. Find a **clockwise** parameterization for an ellipse 10 units wide and 6 units tall (thus with major radius 5 and minor radius 3), centered at the point $(5, 2)$.

Solution. We can parameterize a circle clockwise with $(\sin(t), \cos(t))$ or something else like $(\cos(-t), \sin(-t))$. Thus our ellipse can be

$$(5 \sin(t) + 5, 3 \cos(t) + 2).$$

Alternate solution using the second circle parameterization:

$$(5 \cos(-t) + 5, 3 \sin(-t) + 2).$$